WORKED SOLUTION

STEP I 1990, Question 15

Probability and Statistics • Negative Binomial Distribution

STEP I 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q15

The key idea

For the first part, split the event into two independent pieces: what happens on the last toss, and what happens on all the tosses before it. For the rest, the important fact is that the coin has no memory. Once a head has just appeared, the wait for the next head starts afresh, so we only need to know how likely a geometric waiting time is to be even or odd.

Solution

Write $q = 1 - p$ throughout. Tosses are independent, each a head with probability p.

The distribution of the kth head

Let T_k be the number of the toss on which the kth head appears. The event $\{T_k = n\}$ happens exactly when

- toss n is a head, and
- among tosses $1, 2, \ldots, n-1$ there are exactly $k-1$ heads (and so $n-k$ tails).

Both conditions are needed. If toss n were a tail, the kth head would come before or after toss n, not on it. If the first $n-1$ tosses held a different number of heads, toss n would not give the kth head. Together, the two conditions are also enough.

These two events concern disjoint sets of tosses, so they are independent. The first has probability p. For the second, the $k-1$ heads can sit in any of $\binom{n-1}{k-1}$ sets of positions among the first $n-1$ tosses. Each such arrangement is a particular sequence of $k-1$ heads and $n-k$ tails, with probability $p^{k-1}q^{n-k}$ by independence, and different arrangements are mutually exclusive. Hence

$$\mathrm{P}(T_k = n) = p \cdot \binom{n-1}{k-1} p^{k-1} q^{n-k} = \binom{n-1}{k-1} p^k (1-p)^{n-k}, \qquad n \geq k,$$

as required.

Even to even

Let $G = T_k - T_{k-1}$ be the number of further tosses needed after the $(k-1)$th head (with $T_0 = 0$). The tosses after toss T_{k-1} are independent of everything up to that point. So G is independent of T_{k-1}, and it has the geometric distribution

$$\mathrm{P}(G = g) = q^{g-1}p, \qquad g = 1, 2, \ldots$$

(the case $k = 1$ of the first part).

Ask yourself If T_{k-1} is even, when is T_k even?

Since $T_k = T_{k-1} + G$, if T_{k-1} is even then T_k is even exactly when G is even. Because G is independent of T_{k-1}, conditioning on T_{k-1} being even doesn't change the distribution of G. Therefore

$$\mathrm{P}(T_k \text{ even} \mid T_{k-1} \text{ even}) = \mathrm{P}(G \text{ even}).$$

Summing the geometric series,

$$\mathrm{P}(G \text{ even}) = p\left(q + q^3 + q^5 + \cdots\right) = \frac{pq}{1-q^2} = \frac{pq}{(1-q)(1+q)} = \frac{q}{1+q},$$
$$\mathrm{P}(G \text{ odd}) = p\left(1 + q^2 + q^4 + \cdots\right) = \frac{p}{1-q^2} = \frac{1}{1+q}.$$

So the required probability is

$$\frac{q}{1+q} = \frac{1-p}{2-p}.$$

The answer doesn't depend on k.

Exactly two heads on even-numbered tosses

Now toss until the third head. Let $G_1 = T_1$, $G_2 = T_2 - T_1$ and $G_3 = T_3 - T_2$ be the three waiting times. By the same memoryless argument, they are independent, and each is geometric as above. Write

$$e = \mathrm{P}(G_i \text{ even}) = \frac{q}{1+q}, \qquad o = \mathrm{P}(G_i \text{ odd}) = \frac{1}{1+q}.$$

We start at toss 0, which is even. An even gap keeps the parity of the current position, and an odd gap switches it. The table lists the eight possible parity patterns of (G_1, G_2, G_3) and the resulting parities of (T_1, T_2, T_3):

G_1	G_2	G_3	T_1	T_2	T_3	even heads
E	E	E	E	E	E	3
E	E	O	E	E	O	**2**
E	O	E	E	O	O	1
E	O	O	E	O	E	**2**
O	E	E	O	O	O	0
O	E	O	O	O	E	1
O	O	E	O	E	E	**2**
O	O	O	O	E	O	1

Exactly two heads fall on even tosses in three of the cases, which have probabilities e^2o, eo^2 and o^2e. By independence of the gaps, the required probability is

$$\begin{aligned} e^2o + 2eo^2 = eo(e+2o) &= \frac{q}{(1+q)^2} \cdot \frac{q+2}{1+q} \\ &= \frac{q(2+q)}{(1+q)^3} = \frac{(1-p)(3-p)}{(2-p)^3}. \end{aligned}$$

Checks

- As a check on the first part, summing $\mathrm{P}(T_k = n)$ over $n \geq k$ should give 1. This is the negative binomial series $\sum_{n\geq k} \binom{n-1}{k-1} q^{n-k} = (1-q)^{-k}$.
- The two parity probabilities $\frac{q}{1+q}$ and $\frac{1}{1+q}$ add to 1. When p is close to 1, the first head almost always comes at once, on an odd toss, so $\mathrm{P}(G \text{ even}) \to 0$, as the formula gives.
- For a fair coin ($p = \frac{1}{2}$) the final answer is $\frac{(1/2)(5/2)}{(3/2)^3} = \frac{10}{27} \approx 0.370$. A simulation of 2×10^5 runs gave 0.369. Summing directly over the three waiting times agreed with the formula to 10^{-9} for $p = 0.3$, 0.5 and 0.8.

Common slips

- Using $\binom{n}{k}$ in the first part. The last toss is forced to be a head, so only the first $n-1$ tosses are free.
- Summing the negative binomial over even n in the second part. The memoryless gap avoids this entirely.
- In the last part, counting heads at even positions as if the positions were independent. They aren't, because each position is built from the one before.

Beyond STEP

The time of the kth head has a *negative binomial* distribution. Writing it as a sum of independent geometric gaps is the discrete version of how the arrival times of a Poisson

process are built from independent exponential waits. The parity bookkeeping in the last part is a two-state Markov chain, a topic met early in a first-year probability course. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

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